

Transparency in Debt Crises

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Motivation

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Transparency in emerging market economies

- ▶ Adoption of IMF data transparency reforms [More](#)

Choi and Hashimoto (2018), Gonzalez-Garcia (2022)

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Opaque disclosure in emerging market economies:

- ▶ Greek misreports before the European Debt Crisis.
- ▶ Bolivian reserve reports (2024).
- ▶ Mexico Peso Crisis 1994-1995 [More](#)

When should a sovereign that cannot commit to repay its debt disclose information to creditors?

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The trade-off:

- ▶ By fully disclosing:
 - Lower spreads
 - Issuance according to repayment capacity
- ▶ By not disclosing:
 - more debt at a low spread in bad times.
 - less debt at a high spread in good times

Preview of results

↓ default costs, ↑ incentives not to disclose information ex-ante.

Theory:

- ▶ Non-disclosure functions as **insurance**
- ▶ Optimal under low inefficiencies from default

Hirschleifer (1971) + Grossman and Van Huyck (1988)

Empirics:

- ▶ Higher dispersion of issuances after transparency

Model

Setup

- ▶ Three periods: $t = 0, 1, 2$
- ▶ Continuum of competitive risk-neutral lenders with discount rate $1/R$
- ▶ Sovereign
 - Commits to a disclosure policy $\phi \in \{0, 1\}$ at $t = 0$
 - Has a constant endowment of y , and zero debt due at $t = 1$
 - Cannot commit to repay at $t = 2$
 - Discounts the future at a rate $\beta < 1/R$
- ▶ Endowment after repayment at $t = 2$ can be $i = H, L$ with $\theta_H > \theta_L$:

$$\theta_i = \begin{cases} \theta_H, & \text{with prob. } 1 - \pi \\ \theta_L, & \text{with prob. } \pi \end{cases}$$

- ▶ A sovereign that defaults gets $y_d \leq \theta_L$

Timing

$t = 0$	$t = 1$	$t = 2$
1) Choice of disclosure policy $\phi \in \{0, 1\}$	1) Government learns its type (θ_i) 2) Govt type is revealed ($\phi = 1$) or not ($\phi = 0$) 3) Choice of debt 4) Price is determined q 5) Consumption takes place	1) Default decision 2) Consumption

Budget constraints:

$$c_1(\theta) = y + q(\theta, b(\theta))b(\theta), \quad \forall \theta \quad (t = 1)$$

$$c_2(b, \theta) = \delta(b, \theta) \times y_d + (1 - \delta(b, \theta)) \times (\theta - b), \quad \forall (b, \theta) \quad (t = 2)$$

Type- θ sovereign's problem:

$$V_{FD}(\theta; y_d) = \max_{b(\theta)} \{ \ln(y + q(b, \theta)b(\theta)) + \beta \ln(c_2^*(b, \theta)) \}$$

subject to:

$$c_2^*(b, \theta) = \max_{\delta \in \{0,1\}} \delta \times y_d + (1 - \delta) \times (\theta - b)$$

Investor's profits:

$$\frac{1}{R} [\delta(\theta, b) \times 0 + (1 - \delta(\theta, b))] - q(\theta, b) = 0, \quad \forall (b, \theta)$$

Characterization

Lemma

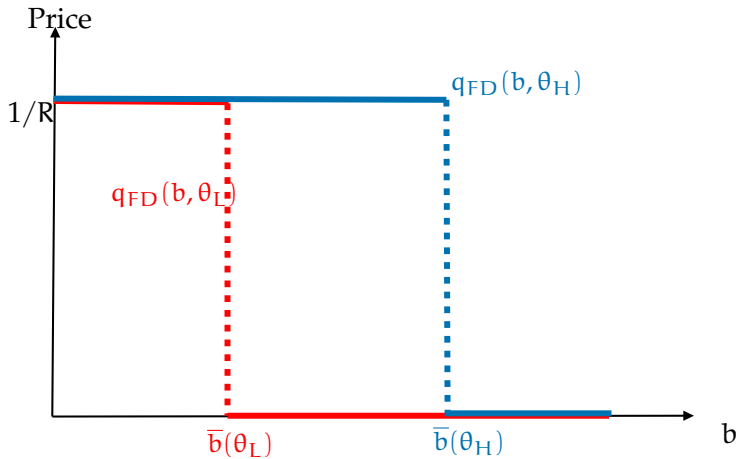
Define $\bar{b}(\theta) \equiv \theta - y_d$. The following strategy and price schedule is an **equilibrium** under full information:

$$\delta_{FD}(b, \theta) = \begin{cases} 1, & \text{if } b > \bar{b}(\theta) \\ 0, & \text{if } b \leq \bar{b}(\theta) \end{cases}$$

$$q_{FD}(b, \theta) = \begin{cases} 0, & \text{if } b > \bar{b}(\theta) \\ 1/R, & \text{if } b \leq \bar{b}(\theta) \end{cases}$$

$$b_{FD}(\theta) = \min \left\{ \frac{\theta - \beta R y}{1 + \beta}, \bar{b}(\theta) \right\}$$

Equilibrium Prices

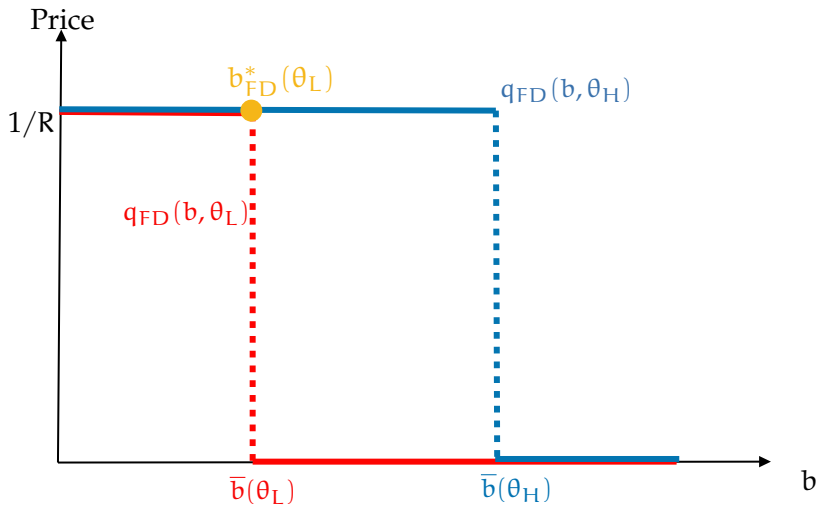


Parametric Assumption

Assumption 1

$$y_d > \frac{\beta R}{1 + \beta} \left(y + \frac{\theta_L}{R} \right)$$

L-type's choice of debt under Assumption 1



Budget constraints:

$$c_1(\theta) = y + q(b(\theta))b(\theta), \quad \forall \theta \quad (t = 1)$$

$$c_2(b, \theta) = \delta(b, \theta) \times y_d + (1 - \delta(b, \theta)) \times (\theta - b), \quad \forall (b, \theta) \quad (t = 2)$$

Type- θ sovereign's problem:

$$V_{ND}(\theta; y_d) = \max_{b(\theta)} \{ \ln(y + q(b(\theta))b(\theta)) + \beta \ln(c_2^*(b, \theta)) \}$$

subject to:

$$c_2^*(b, \theta) = \max_{\delta \in \{0,1\}} \delta \times y_d + (1 - \delta) \times (\theta - b)$$

Investor's expected profits:

$$\frac{1}{R} \mathbb{E}_{\mu} [\delta(b) \times 0 + (1 - \delta(b))] - q(b(\theta)) = 0, \quad \forall b$$

where $\mu(b) \equiv P(\theta = \theta_L | b)$

Proposition

Proof

Under **Assumption 1**, there is no separating debt levels in equilibrium.

Let $\tilde{b}$ be:

$$\tilde{b} \equiv \min \left\{ \frac{\theta_H - \frac{\beta R y}{1-\pi}}{1+\beta}, \theta_H - y_d \right\}$$

Consider the following belief system and default policy:

$$\mu(b) = \begin{cases} \pi, & \text{if } b \leq \tilde{b} \\ 1, & \text{if } b > \tilde{b} \end{cases}$$

$$q_{ND}(b) = \begin{cases} 1/R, & \text{if } b < \theta_L - y_d \\ \frac{1-\mu(b)}{R}, & \text{if } b \in (\theta_L - y_d, \theta_H - y_d] \\ 0, & \text{if } b > \theta_H - y_d \end{cases}$$

$$\delta_{ND}(b, \theta) = \begin{cases} 1, & \text{if } b > \bar{b}(\theta) \\ 0, & \text{if } b \leq \bar{b}(\theta) \end{cases}$$

Risky vs. Safe pooling

Two candidates of pooling equilibrium:

- ▶ *Risky pooling*: both types pick $\tilde{b}$, the L-type defaults.
- ▶ *Safe pooling*: both types pick $\theta_L - y_d$, both types repay.

Define the interval of y_d in which only the L-type is constrained:

$$y_d \in \left(\frac{\beta R}{1 + \beta} \left(y + \frac{\theta_L}{R} \right), \min \left\{ \theta_L, \frac{\beta R}{1 + \beta} \left(y + \frac{\theta_H}{R} \right) \right\} \right) \equiv Y_d$$

Proposition

Proof

Let $y_d \in Y_d$. There exists π^* such that for $\pi < \pi^*$, the unique H-type preferred PBE is the risky pooling equilibrium. Moreover,

$$\frac{\partial \pi^*}{\partial y_d} > 0$$

Disclosure choice

Consider the following condition

$$\log \left(\frac{Ry + \theta_H}{Ry + \theta_L} \right) + \frac{R\beta y - \theta_H}{Ry + \theta_H} > 0 \quad (\star)$$

Define:

$$\Delta(\pi) \equiv V_{ND}(\pi) - V_{FD}(\pi)$$

Proposition

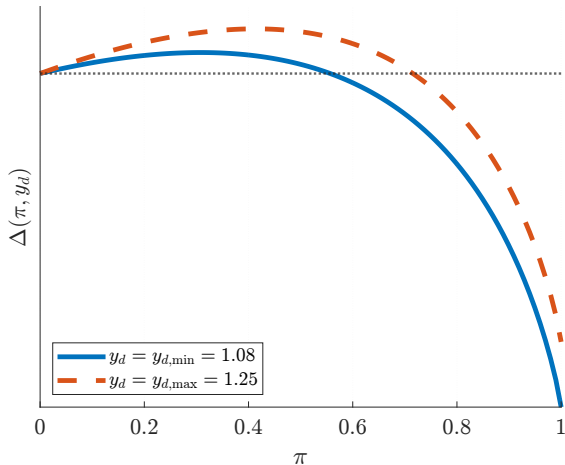
Suppose $(\star)$, holds. Then, for any $y_d \in Y_d$, there exists $\epsilon(y_d) > 0$ such that

$$\Delta(\pi) > 0, \quad \text{for all } \pi \in (0, \epsilon(y_d))$$

That is, non-disclosure (ND) is preferred over full disclosure (FD).

Disclosure choice

- ▶ L-type gains from pooling (**insurance**)
- ▶ H-type loses from pooling (**adverse selection discount**)



Discussion

► Robustness

1. Comparing FD against the whole set of ND [More](#)
2. Types in terms of default endowments [More](#)

► Discussion of assumptions

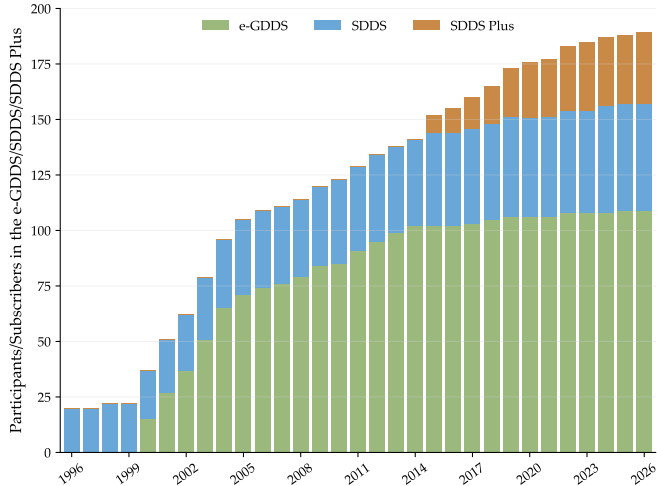
Disclosure policy as a durable rule [More](#)

► Testable implications

Dispersion of debt issuances: transparency vs. opaqueness [More](#)

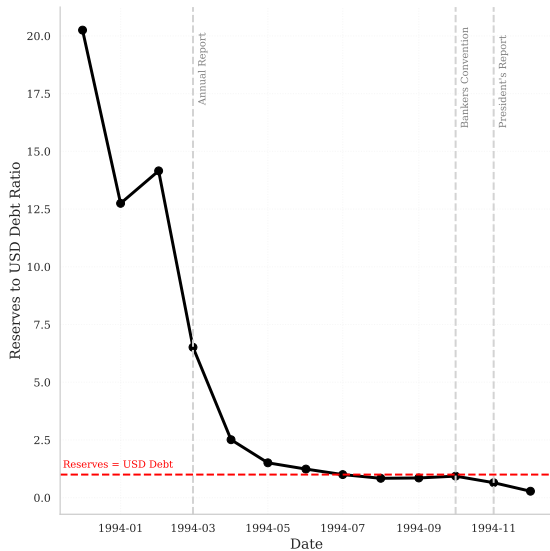
Appendix

History of the Data Standards Initiatives



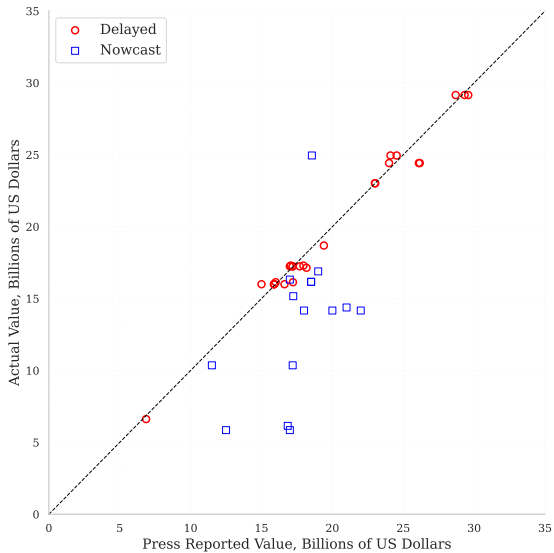
Dollar Reserves vs. Dollar Debt

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Reports vs. Actual Reserve Levels

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1. Sovereign debt under private information

Sandleris (2008), Zabai (2014), Perez (2017), Szkup (2021), Amador and Phelan (2021), [Guler et. al \(2021\)](#), [Horn et. al \(2024\)](#)

2. Role of information in financial crises

Gorton and Ordonez (2020), Faria-e-Castro et. al (2017),

3. Transparency in corporations

Jehiel (2015), Kamenica and Gentzkow (2011), Dye (1983), Verrecchia (1983)

Equilibrium

An **equilibrium under complete information** is a strategy for the sovereign $(\delta_{FD}(\mathbf{b}, \theta), \mathbf{b}_{FD}(\theta))$ and a price schedule for the lenders $\mathbf{q}_{FD}(\mathbf{b}, \theta)$ such that:

- i) The default policy $\delta_{FD}(\theta, \mathbf{b})$ maximizes consumption at $t = 2$.
- ii) Given $\mathbf{q}_{FD}(\mathbf{b}, \theta)$, the choice of debt $\mathbf{b}_{FD}(\theta)$ maximizes the sovereign's lifetime payoff.
- iii) Investors make zero profits.

► Pure strategies (for now)

Definition

A **Perfect Bayesian Equilibrium (PBE)** is a strategy for the sovereign $(\mathbf{b}_{ND}(\theta), \delta_{ND}(\mathbf{b}, \theta))$ and a price schedule for the lender $\mathbf{q}_{ND}(\mathbf{b})$, and beliefs $\mu(\mathbf{b})$ such that:

- 1) The default policy $\delta_{ND}(\mathbf{b}, \theta)$ maximizes the sovereign's consumption at $t = 2$.
- 2) Given $\mathbf{q}_{ND}(\mathbf{b})$ the choice of debt maximizes sovereign's lifetime payoffs.
- 3) Beliefs follow the Bayesian update whenever possible.
- 4) Investors make zero profits.

Definition

A **separating equilibrium** is a PBE in which both types pick different levels of debt: $b_{ND}(\theta_L) \neq b_{ND}(\theta_H)$, so that $\mu(b_{ND}(\theta_L)) = 1$ and $\mu(b_{ND}(\theta_H)) = 0$.

Proof by contradiction:

- ▶ Let $(b_{ND}(\theta), \delta_{ND}(b, \theta), q_{ND}(b), \mu(b))$ be a separating equilibrium.
- ▶ By definition $b_{ND}(\theta_L) \neq b_{ND}(\theta_H)$, with $\mu(b_{ND}(\theta_L)) = 1$ and $\mu(b_{ND}(\theta_H)) = 0$.
- ▶ Since investors know with which type they are dealing with prices are $q(b) \in \{0, 1/R\}$ for $b \in \{b_{ND}(\theta_L), b_{ND}(\theta_H)\}$, due to the zero profit condition.
- ▶ Note that regardless of μ , $(\theta_L - y_d, 1/R)$ is always available for both types.
- ▶ Therefore, it must be the case that $q(b) = 1/R$ for $b \in \{b_{ND}(\theta_L), b_{ND}(\theta_H)\}$.
- ▶ It then follows from **Assumption 1** that $b_{ND}(\theta_L) = \theta_L - y_d$.
- ▶ Since $b_{ND}(\theta_H) < \theta_L - y_d$ is not optimal for the H-type, and $q(b_{ND}(\theta_H)) = 1/R$, it must be that $\theta_H - y_d \geq b_{ND}(\theta_H) > \theta_L - y_d$.
- ▶ But this is a profitable deviation for the L-type, that would prefer to mimic, therefore $(b_{ND}(\theta), \delta_{ND}(b, \theta), q_{ND}(b), \mu(b))$ cannot be a separating equilibrium.

Definition

A **pooling equilibrium** is a PBE in which both types pick the same debt level:

$$b_{ND}(\theta_L) = b_{ND}(\theta_H) = \tilde{b}$$

Best Default Outcome

$$\max_{\{\tilde{b}, \tilde{q}\}} u(y + \tilde{q}\tilde{b}) + \beta u(\theta_H - \tilde{b}) \quad (P)$$

subject to

$$\tilde{b} \leq \theta_H - y_d \quad (\text{SUST-H}, \nu)$$

$$\tilde{b} \geq \theta_L - y_d \quad (\text{ICD-L}, \chi)$$

$$u(y + \tilde{q}\tilde{b}) + \beta u(y_d) \geq u(y) + \beta u(\theta_L) \quad (\text{PC-L}, \eta)$$

$$\underbrace{\pi \times [-\tilde{q}\tilde{b}] + (1 - \pi) \times \left[\frac{\tilde{b}}{R} - \tilde{q}\tilde{b} \right]}_{\frac{(1-\pi)}{R} \tilde{b} - \tilde{q}\tilde{b}} \geq 0 \quad (\text{PC-I}, \xi)$$

Lemma

Suppose $y_d \leq \theta_L$, and the L-type defaults, that is, $\tilde{b} > \theta_L - y_d$ ($\chi = 0$). Therefore, investors make zero profits ($\xi > 0$), that is,

$$\tilde{q} = \frac{1 - \pi}{R}$$

Proof. Let $(\tilde{q}, \tilde{b})$ be a solution to (P). Assume, by contradiction, that investors make positive profits, that is, $\tilde{q} < \frac{1-\pi}{R}$. Since the H-type's payoff is strictly increasing in $\tilde{q}$, it must be the case that there exists $q' > \tilde{q}$ that increases the H-type's payoff. Since the payoff of type L is also increasing in the price of the debt, and when $\tilde{q}$ PC-L is satisfied, we must have that q' also satisfies PC-L, which means that $\tilde{q}$ is not a solution to (P).

The risky pooling contract maximizes the payoff of the H-type among all contracts in which the L-type defaults.

Proof.
$$(1 + \eta)u' \left(y + \frac{1 - \pi}{R} \times \tilde{b} \right) (1 - \pi) - R\beta u'(\theta_H - \tilde{b}) = R(v - \chi) \quad (1)$$

By looking for a solution in which the L-type, defaults, that is, $\tilde{b} \in (\theta_L - y_d, \theta_H - y_d)$, we have that $v = \chi = 0$. If the L-type's PC-I binds but PC-L is slack ($\eta = 0$), we have:

$$\tilde{b} = \frac{\theta_H - \frac{R\beta}{1 - \pi} \times y}{1 + \beta}$$

We now look for a parametrization in which $\eta = 0$. Notice that if $y_d = \theta_L$, PC-L is slack. By continuity of $u \left(y + \frac{1 - \pi}{R} \tilde{b} \right) + \beta u(y_d)$ in y_d , there must exist $\epsilon > 0$ such that PC-L holds with strict inequality for $y_d \in (\theta_L - \epsilon, \theta_L)$. Define:

$$\bar{\epsilon} \equiv \sup\{\epsilon : \epsilon > 0, \text{ such that PC-L holds with strict inequality for } y_d = \theta_L - \epsilon\}$$

By taking $y_d \equiv \theta_L - \bar{\epsilon}$ we have that $\eta = 0$.

Best Repayment Outcome

$$\max_{\{b_H, b_L, q_H, q_L\}} u(y + q_H b_H) + \beta u(\theta_H - b_H) \quad (S)$$

subject to

$$b_H \leq \theta_H - y_d \quad (\text{SUST-H}, v_H)$$

$$b_L \leq \theta_L - y_d \quad (\text{SUST-L}, v_L)$$

$$u(y + q_L b_L) + \beta u(\theta_L - b_L) \geq u(y + q_H b_H) + \beta u(y_d) \quad (\text{ICD-L}, \chi)$$

$$u(y + q_L b_L) + \beta u(\theta_L - b_L) \geq u(y + q_H b_H) + \beta u(\theta_L - b_H) \quad (\text{ICR-L}, \lambda)$$

$$u(y + q_L b_L) + \beta u(\theta_L - b_L) \geq v_L^a \quad (\text{PC-L}, \eta)$$

$$b_L \times \left[\frac{1}{R} - q_L \right] \geq 0 \quad (\text{PCI-L}, \xi_L)$$

$$b_H \times \left[\frac{1}{R} - q_H \right] \geq 0 \quad (\text{PCI-H}, \xi_H)$$

Proposition

The safe pooling outcome maximizes the H-type's payoff in which: (i) lenders make zero profits; (ii) the sustainability constraint for the L-type binds; (iii) the L-type prefers to default rather than repay the H-type's contract.

Proof.

We look for a solution with $\xi_H, \xi_L > 0$, which implies $q_H = q_L = 1/R$. By the FOC with respect to (b_H) :

$$u' \left(y + \frac{b_L}{R} \right) - R\beta u'(\theta_L - b_L) = \frac{Rv_L}{\chi + \lambda + \eta}$$

Consider first the case in which the sustainability constraint binds for the L-type: $v_L > 0$, that is, $b_L = \theta_L - y_d$. We can then rewrite the ICD-L as:

$$u\left(y + \frac{\theta_L - y_d}{R}\right) \geq u\left(y + \frac{b_H}{R}\right) \implies b_H \leq \theta_L - y_d$$

Since SUST-L binds we know that the L-type is indifferent between default and repayment of b_L . By the FOC with respect to b_H :

$$u'(y + b_H/R) - R\beta u'(\theta_H - b_H) = \chi u'(y + b_H/R) + \lambda[u'(y + b_H/R) - R\beta u'(\theta_L - b_H)]$$

If the relevant constraint is ICD-L (picking the H-type's contract and defaulting is better than picking the H-type's contract and repaying) it must be the case that:

$$u(y_d) \geq u(\theta_L - b_H)$$

That is:

$$b_H \geq \theta_L - y_d$$

Only L-type is constrained:

$$y_d \in \left(\frac{\beta R}{1 + \beta} \left(y + \frac{\theta_L}{R} \right), \frac{\beta R}{1 + \beta} \left(y + \frac{\theta_H}{R} \right) \right)$$

Lemma

There exists $\pi_L(y_d)$ such that, for $\pi < \pi_L(y_d)$, the L-type prefers the *risky pooling debt level*

Proof.

$$y + \frac{1 - \pi}{R} \left[\frac{\theta_H - \frac{\beta R y}{1 - \pi}}{1 + \beta} \right] > y + \frac{\theta_L - y_d}{R}$$

$$\underbrace{1 - \left[\frac{(1 + \beta)(\theta_L - y_d) + \beta R y}{\theta_H} \right]}_{\equiv \pi_L(y_d)} > \pi$$

Define the risky pooling payoff for the H-type:

$$\mathcal{F}(\pi) \equiv (1 + \beta) \log \left(\frac{y + \frac{1-\pi}{R} \times \theta_H}{1 + \beta} \right) + \beta \log \frac{\beta R}{1 - \pi}$$

Lemma

$\mathcal{F}$ is strictly convex. Moreover, at its minimum ρ' :

$$\mathcal{F}(\rho') = \log(y) + \beta \log(\theta_H)$$

Lemma

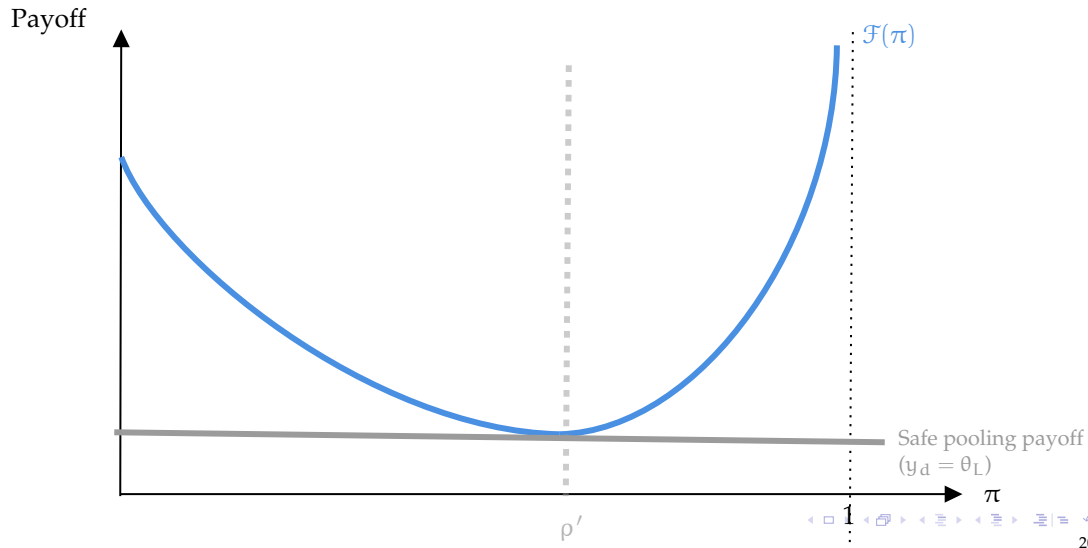
There exists $\pi_H(y_d)$ such that, for $\pi < \pi_H(y_d)$, the H-type prefers the *risky pooling* debt level

Proposition

Suppose only the L-type is constrained. There exists π^* such that for $\pi < \pi^*$, the unique pooling equilibrium has $\tilde{b} = \frac{\theta_H - \frac{\beta R y}{1-\pi}}{1+\beta}$

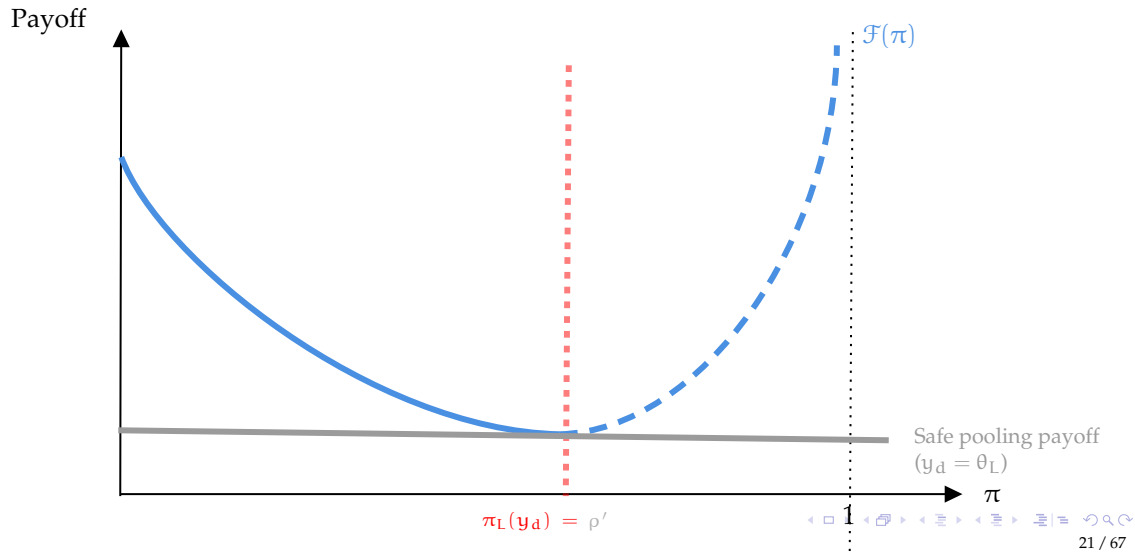
Pooling Eq. - General Case

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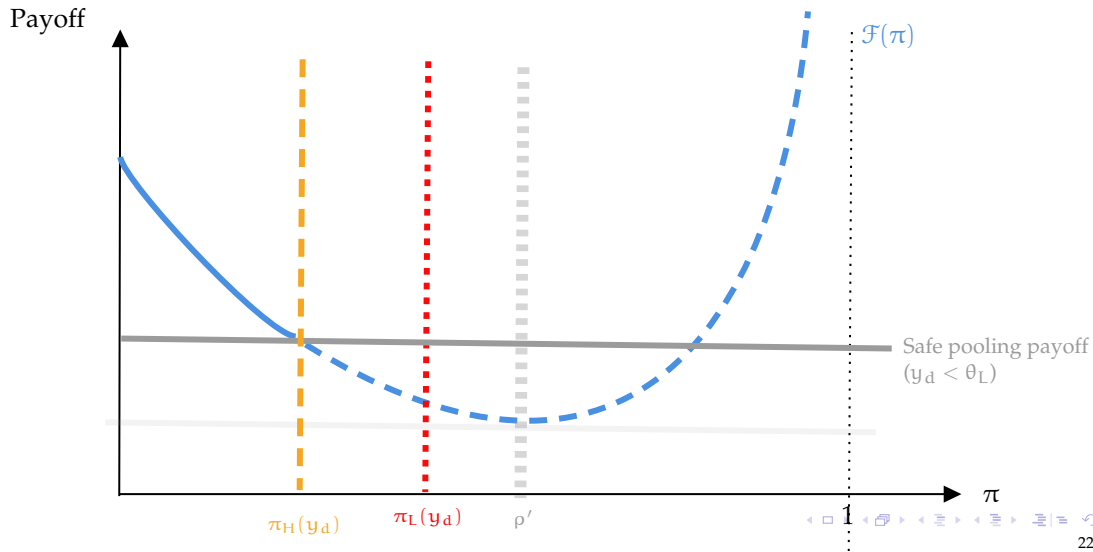
Pooling Eq. - General Case

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Pooling Eq. - General Case

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Robustness

Compare the FD vs. the whole set of ND

- Set of ND equilibrium

$$\mathcal{E}_{\text{ND}}(\pi, y_d) \equiv \{\text{all pooling PBEs under no disclosure}\}$$

- Pooling equilibrium in our model is $e = (b, \delta(b), q_{\text{ND}}(b), \mu(b))$

$$\bar{V}_{\text{ND}}(\pi, y_d) \equiv \sup_{e \in \mathcal{E}_{\text{ND}}(\pi, y_d)} V_0(e), \quad \underline{V}_{\text{ND}}(\pi, y_d) \equiv \inf_{e \in \mathcal{E}_{\text{ND}}(\pi, y_d)} V_0(e)$$

L-type's IC.

$$v_L^R(b) = u\left(y + \frac{1-\pi}{R}b\right) + \beta u(y_d) \geq u\left(y + \frac{\theta_L - y_d}{R}\right) + \beta u(y_d) = v_L^S$$

$$b \geq \frac{\theta_L - y_d}{1-\pi}$$

H-type's IC.

$$v_H^R(b) = u\left(y + \frac{1-\pi}{R}b\right) + \beta u(\theta_H - b) \geq u\left(y + \frac{\theta_L - y_d}{R}\right) + \beta u(\theta_H - \theta_L + y_d) = v_H^S$$

$$v_H^R(b) \geq v_H^S$$

Define the set of risky pooling debt levels:

$$\mathcal{B}^R(\pi, y_d) \equiv \left\{ b \in (\theta_L - y_d, \theta_H - y_d] : b \geq \frac{\theta_L - y_d}{1-\pi}, \quad v_H^R \geq v_H^S \right\}$$

Lemma

Under ND, we must have that

$$\underline{V}_{\text{ND}}(\pi, y_d) = V^S(\pi, y_d)$$

Proof.

Notice that any *risky-pooling* satisfies the IC constraints:

$$v_H^R(b) \geq v_H^S$$

$$v_L^R(b) \geq v_L^S$$

which means that any risky-pooling is at least as good as the safe debt level without default.

Lemma

The safe pooling payoff under ND is dominated by FD.

$$\underline{V}_{ND}(\pi, y_d) < V_{FD}(\pi, y_d)$$

Proof.

We should start noticing that the L-type is indifferent between the safe pooling and FD since $b_{FD}^*(\theta_L) = \theta_L - y_d$. Under the **Assumption 1**, however, the H-type's solution under FD lies in the risky region. Which means that the H-type is strictly better off under FD. $\square$

$$\max_{b \in \mathcal{B}_{ND}(\pi, y_d)} \log \left(y + \frac{1-\pi}{R} b \right) + \beta \left[\pi \log(y_d) + (1-\pi) \log(\theta_H - b) \right]$$

Taking FOCs:

$$\hat{b}(\pi) = \frac{\theta_H - \beta R y}{1 + \beta(1 - \pi)}$$

Notice that

$$\forall \pi > 0, \quad \hat{b}(\pi) > \tilde{b}(\pi)$$

$\Rightarrow$ Ex-ante optimal debt level puts weight on the L-type
(who prefers more debt)

Proposition

There exists $\hat{\pi}(y_d) > 0$ for any $y_d \in Y_d$, such that for $\pi < \hat{\pi}(y_d)$, the debt level that maximizes ex-ante welfare is a risky-pooling PBE, that is, $\hat{b}(\pi) \in \mathcal{B}^{ND}(\pi, y_d)$

Lemma

Let

$$\hat{\pi}_C(y_d) \equiv \frac{\frac{\beta R}{1+\beta} \left(y + \frac{\theta_H}{R} \right) - y_d}{\frac{\beta}{1+\beta} (\theta_H - y_d)}$$

If $\pi < \hat{\pi}_C(y_d)$, then $\hat{b}(\pi) \in (\theta_L - y_d, \theta_H - y_d]$ for each $y_d \in Y_d$.

Proof: Notice that:

$$b'_W(\pi) = \frac{(\theta_H - \beta R y) \beta}{[1 + \beta(1 - \pi)]^2} > 0 \quad (\text{since } \theta_H > \beta R y \text{ is necessary for the H-type to borrow})$$

We also have that:

$$\begin{aligned} \hat{b}(0) &= \frac{\theta_H - \beta R y}{1 + \beta} > \theta_L - y_d \\ \theta_H - \beta R y &> \theta_L + \beta \theta_L - (1 + \beta) y_d \\ (1 + \beta) y_d &> \theta_L - \theta_H + \beta \theta_L + \beta R y \\ y_d &> \underbrace{\frac{\theta_L - \theta_H}{1 + \beta}}_{< 0} + \frac{\beta R}{1 + \beta} \left(y + \frac{\theta_L}{R} \right) \end{aligned}$$

We then conclude that $\hat{b}(\pi) > \theta_L - y_d$ for any $\pi \in [0, 1]$ and $y_d \in Y_d$.

We now need to show that:

$$\begin{aligned}
 \hat{b}(\pi) \leq \theta_H - y_d &\iff \frac{\theta_H - \beta R y}{1 + \beta(1 - \pi)} \leq \theta_H - y_d, \\
 &\iff \theta_H - \beta R y \leq (1 + \beta(1 - \pi))(\theta_H - y_d), \\
 &\iff \theta_H - \beta R y \leq (1 + \beta)\theta_H - (1 + \beta)y_d - \beta\pi(\theta_H - y_d), \\
 &\iff \beta\pi(\theta_H - y_d) \leq \beta(Ry + \theta_H) - (1 + \beta)y_d, \\
 &\iff \pi \leq \frac{\beta(Ry + \theta_H) - (1 + \beta)y_d}{\beta(\theta_H - y_d)} \\
 &\iff \pi \leq \frac{\overbrace{\frac{\beta R}{1 + \beta} \left(y + \frac{\theta_H}{R} \right) - y_d}^{>0}}{\frac{\beta}{1 + \beta}(\theta_H - y_d)} \equiv \hat{\pi}_C(y_d)
 \end{aligned}$$

Lemma

There exists $\hat{\pi}_L(y_d) > 0$ for any $y_d \in Y_d$ such that, for $\pi < \hat{\pi}_L(y_d)$ the L-types IC holds.

Proof.

$$v_L^R(\pi) = \log \left(y + \frac{1-\pi}{R} \hat{b}(\pi) \right) + \beta \log(y_d) \geq \log \left(y + \frac{\theta_L - y_d}{R} \right) + \beta \log(y_d)$$

$$\begin{aligned}
(1 - \pi)\hat{b}(\pi) &\geq \theta_L - y_d && \iff (1 - \pi)\frac{\theta_H - \beta Ry}{1 + \beta(1 - \pi)} \geq \theta_L - y_d, \\
&&& \iff (1 - \pi)(\theta_H - \beta Ry) \geq (1 + \beta(1 - \pi))(\theta_L - y_d), \\
&&& \iff (1 - \pi)(\theta_H - \beta Ry) \geq (\theta_L - y_d) + \beta(1 - \pi)(\theta_L - y_d), \\
&&& \iff (1 - \pi)[\theta_H - \beta Ry - \beta(\theta_L - y_d)] \geq \theta_L - y_d, \\
&&& \iff 1 - \pi \geq \frac{\theta_L - y_d}{\theta_H - \beta Ry - \beta(\theta_L - y_d)}, \\
&&& \iff \pi \leq 1 - \frac{\theta_L - y_d}{\theta_H - \beta Ry - \beta(\theta_L - y_d)}, \\
&&& \iff \pi \leq \frac{\overbrace{\theta_H - \theta_L}^{>0} + \overbrace{(1 + \beta)y_d - \beta(Ry + \theta_L)}^{>0}}{\underbrace{\theta_H - y_d}_{>0} + \underbrace{(1 + \beta)y_d - \beta(Ry + \theta_L)}_{>0}} \equiv \hat{\pi}_L(y_d).
\end{aligned}$$

Notice that $(1 + \beta)y_d > \beta(Ry + \theta_L)$. And since $y_d \leq \theta_L$, we have that

Lemma

There exists $\hat{\pi}_L(y_d) > 0$ for any $y_d \in Y_d$ such that, for $\pi < \hat{\pi}_L(y_d)$ the L-types IC holds.

Proof.

$$u\left(y + \frac{1-\pi}{R}\hat{b}(\pi)\right) + \beta u(\theta_H - \hat{b}(\pi)) \geq u\left(y + \frac{\theta_L - y_d}{R}\right) + \beta u(\theta_H - \theta_L + y_d) = v_H^S$$

Notice that:

$$\begin{aligned}
 y + \frac{1-\pi}{R} \times \frac{\theta_H - \beta Ry}{1 + \beta(1-\pi)} &= \frac{Ry + \beta Ry(1-\pi) + (1-\pi)\theta_H - (1-\pi)\beta Ry}{Ry[1 + \beta(1-\pi)]} \\
 &= \frac{Ry + (1-\pi)\theta_H}{Ry[1 + \beta(1-\pi)]} \\
 &= \frac{y + \frac{1-\pi}{R}\theta_H}{y[1 + \beta(1-\pi)]} \\
 \theta_H - \frac{\theta_H - \beta Ry}{1 + \beta(1-\pi)} &= \frac{\theta_H + \theta_H\beta(1-\pi) - \theta_H + \beta Ry}{1 + \beta(1-\pi)} \\
 &= \frac{\beta[\theta_H(1-\pi) + Ry]}{1 + \beta(1-\pi)} \\
 &= \frac{R\beta}{1 + \beta(1-\pi)} \left(y + \frac{1-\pi}{R}\theta_H \right)
 \end{aligned}$$

Hence the H-type's incentive-compatibility constraint is

$$\begin{aligned} \log(\hat{c}_{1H}(\pi)) + \beta \log(\hat{c}_{2H}(\pi)) &\geq \log\left(y + \frac{\theta_L - y_d}{R}\right) + \beta \log(\theta_H - \\ \log\left(\frac{Ry + (1 - \pi)\theta_H}{R(1 + \beta(1 - \pi))}\right) + \beta \log\left(\frac{\beta(Ry + (1 - \pi)\theta_H)}{1 + \beta(1 - \pi)}\right) &\geq \log\left(\frac{Ry + \theta_L - y_d}{R}\right) + \beta \log(\theta_H - \\ (1 + \beta) \log\left(\frac{Ry + (1 - \pi)\theta_H}{1 + \beta(1 - \pi)}\right) + \beta \log \beta &\geq \log(Ry + \theta_L - y_d) + \beta \log(\theta_H - \\ \frac{Ry + (1 - \pi)\theta_H}{1 + \beta(1 - \pi)} &\geq \left[\frac{(Ry + \theta_L - y_d)(\theta_H - \theta_L + y_d)}{\beta^\beta} \right] \end{aligned}$$

Define

$$\mathcal{F}(\pi) \equiv \frac{Ry + (1 - \pi)\theta_H}{1 + \beta(1 - \pi)}.$$

Then the H-type's incentive-compatibility constraint is equivalent to

$$\mathcal{F}(\pi) \geq \left[\frac{(Ry + \theta_L - y_d)(\theta_H - \theta_L + y_d)^\beta}{\beta^\beta} \right]^{\frac{1}{1+\beta}}.$$

Moreover,

$$\begin{aligned} \mathcal{F}'(\pi) &= \frac{-\theta_H(1 + \beta(1 - \pi)) + \beta(Ry + (1 - \pi)\theta_H)}{(1 + \beta(1 - \pi))^2} \\ &= \frac{\beta Ry - \theta_H}{(1 + \beta(1 - \pi))^2}. \end{aligned}$$

Therefore, whenever $\theta_H > \beta Ry$,

$$\mathcal{F}'(\pi) < 0,$$

At $\pi = 0$,

$$\hat{b}(0) = \frac{\theta_H - \beta R y}{1 + \beta} = b_{FD}^*(\theta_H).$$

Since $y_d \in Y_d$, the H-type is unconstrained at the risk-free price, and

$$\hat{b}(0) = b_{FD}^*(\theta_H) > \theta_L - y_d.$$

Because

$$b \mapsto \log\left(y + \frac{b}{R}\right) + \beta \log(\theta_H - b)$$

is strictly concave and uniquely maximized at $\hat{b}(0)$, it follows that

$$\log\left(y + \frac{\hat{b}(0)}{R}\right) + \beta \log(\theta_H - \hat{b}(0)) > \log\left(y + \frac{\theta_L - y_d}{R}\right) + \beta \log(\theta_H - \theta_L + y_d).$$

Equivalently,

$$\mathcal{F}(0) > \left[\frac{(\mathbf{R}\mathbf{y} + \theta_L - \mathbf{y}_d)(\theta_H - \theta_L + \mathbf{y}_d)^\beta}{\beta^\beta} \right]^{\frac{1}{1+\beta}}.$$

Since $\mathcal{F}$ is continuous and strictly decreasing, there exists a unique cutoff $\hat{\pi}_H(\mathbf{y}_d) \in (0, 1]$ such that the H-type's incentive-compatibility constraint holds if and only if

$$\pi \leq \hat{\pi}_H(\mathbf{y}_d).$$

If

$$\left[\frac{(\mathbf{R}\mathbf{y} + \theta_L - \mathbf{y}_d)(\theta_H - \theta_L + \mathbf{y}_d)^\beta}{\beta^\beta} \right]^{\frac{1}{1+\beta}} > \mathbf{R}\mathbf{y},$$

then $\hat{\pi}_H(\mathbf{y}_d)$ is characterized by

$$\frac{\mathbf{R}\mathbf{y} + (1 - \hat{\pi}_H(\mathbf{y}_d))\theta_H}{1 + \beta(1 - \hat{\pi}_H(\mathbf{y}_d))} = \left[\frac{(\mathbf{R}\mathbf{y} + \theta_L - \mathbf{y}_d)(\theta_H - \theta_L + \mathbf{y}_d)^\beta}{\beta^\beta} \right]^{\frac{1}{1+\beta}},$$

that is,

$$\hat{\pi}_H(y_d) = 1 - \frac{\left[\frac{(Ry + \theta_L - y_d)(\theta_H - \theta_L + y_d)^\beta}{\beta^\beta} \right]^{\frac{1}{1+\beta}} - Ry}{\theta_H - \beta \left[\frac{(Ry + \theta_L - y_d)(\theta_H - \theta_L + y_d)^\beta}{\beta^\beta} \right]^{\frac{1}{1+\beta}}}.$$

If instead

$$\left[\frac{(Ry + \theta_L - y_d)(\theta_H - \theta_L + y_d)^\beta}{\beta^\beta} \right]^{\frac{1}{1+\beta}} \leq Ry,$$

then the H-type's incentive-compatibility constraint holds for all $\pi \in [0, 1)$, so

$$\hat{\pi}_H(y_d) = 1.$$

Robustness

Types in terms of default costs

Complete Information Benchmark

- Types are described with respect to the deadweight losses from default

$$0 < \theta_L < \theta_H \quad (\text{L-types more tempted to default})$$

- Consumption

$$c_1 = y_1 + qb$$

$$c_2^R = y_2 - b$$

$$c_2^D(\theta) = y_2 - \theta$$

- Default decision

$$y_2 - \theta \geq y_2 - b$$

$$b \leq \theta$$

Equilibrium under complete information

$$\delta^*(b, \theta) = \begin{cases} 0, & b \leq \theta \\ 1, & b > \theta \end{cases} \quad (2)$$

$$q_{FD}(b, \theta) = \begin{cases} 1/R, & b \leq \theta \\ 0, & b > \theta \end{cases} \quad (3)$$

$$b_{FD}^*(\theta) = \min \left\{ \frac{y_2 - R\beta y_1}{1 + \beta}, \theta \right\} \quad (4)$$

The above equation implies that $b_{FD}^*(\theta_H) > b_{FD}^*(\theta_L)$. We now will impose the equivalent of **Assumption 1** (to ensure only the L-type is constrained).

$$\theta_L < \frac{y_2 - R\beta y_1}{1 + \beta} \leq \theta_H$$

Incomplete Information

Proposition

If

$$\theta_L < \frac{y_2 - R\beta y_1}{1 + \beta} \leq \theta_H,$$

then there is no separating PBE

Safe Pooling

$$b = \theta_L$$

The ex-ante payoff is then:

$$\underline{V}_{ND} = \log \left(y_1 + \frac{\theta_L}{R} \right) + \beta [\pi \log(y_2 - \theta_L) + (1 - \pi) \log(y_2 - \theta_L)]$$

$$\underline{V}_{ND} = \log \left(y_1 + \frac{\theta_L}{R} \right) + \beta \log(y_2 - \theta_L)$$

Since the H-type is not constrained under complete information we have that

Lemma

The full-disclosure equilibrium dominates the safe-pooling equilibrium.

$$V_{FD}(\pi) > \underline{V}_{ND}, \quad \forall \pi \in [0, 1]$$

Risky Pooling

For the risky pooling to be sustained, we need it to satisfy both types's ICs. For the L-type:

$$\begin{aligned}\log\left(y + \frac{1-\pi}{R}b\right) + \beta \log(y_2 - \theta_L) &\geq \log\left(y + \frac{\theta_L}{R}\right) + \beta \log(y_2 - \theta_L) \\ (1-\pi)b &\geq \theta_L \\ b &\geq \frac{\theta_L}{1-\pi}\end{aligned}$$

But recall that for b to be in the risky region $b \leq \theta_H$, so:

$$\begin{aligned}\frac{\theta_L}{1-\pi} &\leq \theta_H \\ \pi &\leq 1 - \frac{\theta_L}{\theta_H}\end{aligned}$$

For the H-type

$$\hat{v}_H(b; \pi) = \log \left(y_1 + \frac{1-\pi}{R} b \right) + \beta \log(y_2 - b)$$

to

$$v_H^{SP} = \log \left(y_1 + \frac{\theta_L}{R} \right) + \beta \log(y_2 - \theta_L).$$

Then the H-type IC is

$$\hat{v}_H(b; \pi) \geq v_H^{SP}$$

Combining the two incentive constraints, the set of risky-pooling equilibria is

$$\mathcal{B}^{ND}(\pi) = \left\{ b \in (\theta_L, \theta_H] : b \geq \frac{\theta_L}{1-\pi}, \hat{v}_H(b; \pi) \geq v_H^{SP} \right\}.$$

Ex-ante optimal debt level

Ex-ante welfare is

$$\hat{V}(b; \pi) = \log \left(y_1 + \frac{1-\pi}{R} b \right) + \beta \left[\pi \log(y_2 - \theta_L) + (1-\pi) \log(y_2 - b) \right].$$

Taking FOCs and solving for the optimal debt:

$$\begin{aligned} \frac{1}{Ry_1 + (1-\pi)b} &= \frac{\beta}{y_2 - b}, \\ y_2 - b &= \beta(Ry_1 + (1-\pi)b), \\ y_2 - \beta Ry_1 &= (1 + \beta(1-\pi))b. \\ \hat{b}(\pi) &= \frac{y_2 - \beta Ry_1}{1 + \beta(1-\pi)}. \end{aligned}$$

Define

$$\bar{V}_{ND}(\pi) \equiv \hat{V}(\hat{b}(\pi); \pi).$$

Substituting $\hat{b}(\pi)$, period-1 pooled consumption is

$$\begin{aligned}\hat{c}_1(\pi) &= y_1 + \frac{1-\pi}{R} \hat{b}(\pi) \\ &= y_1 + \frac{1-\pi}{R} \frac{y_2 - \beta R y_1}{1 + \beta(1-\pi)} \\ &= \frac{R y_1 + (1-\pi) y_2}{R(1 + \beta(1-\pi))},\end{aligned}$$

while type H's period-2 consumption is

$$\begin{aligned}\hat{c}_{2H}(\pi) &= y_2 - \hat{b}(\pi) \\ &= y_2 - \frac{y_2 - \beta R y_1}{1 + \beta(1-\pi)} \\ &= \frac{\beta(R y_1 + (1-\pi) y_2)}{1 + \beta(1-\pi)}.\end{aligned}$$

Hence

$$\begin{aligned}\bar{V}_{\text{ND}}(\pi) = & \log \left(\frac{Ry_1 + (1 - \pi)y_2}{R(1 + \beta(1 - \pi))} \right) + \beta\pi \log(y_2 - \theta_L) \\ & + \beta(1 - \pi) \log \left(\frac{\beta(Ry_1 + (1 - \pi)y_2)}{1 + \beta(1 - \pi)} \right).\end{aligned}$$

We now show that $\hat{b}(\pi)$ can be sustained as a risky-pooling equilibrium for sufficiently small π . This requires:

1. $\hat{b}(\pi) \in (\theta_L, \theta_H]$,
2. the low type does not deviate to safe pooling at $b = \theta_L$,
3. the high type does not deviate to safe pooling at $b = \theta_L$.

Lemma

If

$$\theta_L < \frac{y_2 - \beta R y_1}{1 + \beta} \leq \theta_H,$$

There exists $\hat{\pi}_C$ such that for any $\pi < \hat{\pi}_C$, $\hat{b}(\pi) \in (\theta_L, \theta_H]$.

Proof.

In our region of interest:

$$\hat{b}(0) = \frac{y_2 - \beta R y_1}{1 + \beta} > \theta_L.$$

Also,

$$\hat{b}'(\pi) = \frac{\beta(y_2 - \beta R y_1)}{(1 + \beta(1 - \pi))^2} > 0,$$

provided $y_2 > \beta R y_1$.

Hence

$$\hat{b}(\pi) > \theta_L \quad \text{for all } \pi \in [0, 1).$$

For the upper end of the risky region, we need

$$\begin{aligned} \hat{b}(\pi) \leq \theta_H & \iff \frac{y_2 - \beta R y_1}{1 + \beta(1 - \pi)} \leq \theta_H \\ & \iff y_2 - \beta R y_1 \leq \theta_H (1 + \beta(1 - \pi)) \\ & \iff \pi \leq \frac{(1 + \beta)\theta_H - y_2 + \beta R y_1}{\beta \theta_H} \equiv \hat{\pi}_C. \end{aligned}$$

Lemma

There exists $\hat{\pi}_L > 0$ such that, for any $\pi < \hat{\pi}_L$ the L-type's IC is satisfied.

Proof.

Under risky pooling at $\hat{b}(\pi)$, the low type gets

$$\log \left(y_1 + \frac{1-\pi}{R} \hat{b}(\pi) \right) + \beta \log(y_2 - \theta_L),$$

while under safe pooling at $b = \theta_L$, it gets

$$\log \left(y_1 + \frac{\theta_L}{R} \right) + \beta \log(y_2 - \theta_L).$$

Therefore the low-type IC is

$$\begin{aligned}(1 - \pi)\hat{b}(\pi) &\geq \theta_L &\iff (1 - \pi)\frac{y_2 - \beta Ry_1}{1 + \beta(1 - \pi)} &\geq \theta_L \\ &&\iff (1 - \pi)(y_2 - \beta Ry_1) &\geq \theta_L(1 + \beta(1 - \pi)) \\ &&\iff (1 - \pi)(y_2 - \beta Ry_1 - \beta\theta_L) &\geq \theta_L \\ &&\iff \pi \leq \frac{y_2 - \beta Ry_1 - (1 + \beta)\theta_L}{y_2 - \beta Ry_1 - \beta\theta_L} &\equiv \hat{\pi}_L.\end{aligned}$$

Under the interesting-region assumption, $\hat{\pi}_L > 0$.

Lemma

There exists $\hat{\pi}_H > 0$ such that for any $\pi < \hat{\pi}_H$ the H-type's IC hold.

Proof. Under risky pooling at $\hat{b}(\pi)$, the high type gets

$$\log(\hat{c}_1(\pi)) + \beta \log(\hat{c}_{2H}(\pi)) = (1 + \beta) \log\left(\frac{Ry_1 + (1 - \pi)y_2}{1 + \beta(1 - \pi)}\right) + \beta \log \beta - \log R.$$

Under safe pooling, it gets

$$\log\left(y_1 + \frac{\theta_L}{R}\right) + \beta \log(y_2 - \theta_L) = \log(Ry_1 + \theta_L) - \log R + \beta \log(y_2 - \theta_L).$$

Hence the high-type IC is

$$(1 + \beta) \log\left(\frac{Ry_1 + (1 - \pi)y_2}{1 + \beta(1 - \pi)}\right) + \beta \log \beta \geq \log(Ry_1 + \theta_L) + \beta \log(y_2 - \theta_L).$$

Equivalently,

$$\frac{Ry_1 + (1 - \pi)y_2}{1 + \beta(1 - \pi)} \geq \left[\frac{(Ry_1 + \theta_L)(y_2 - \theta_L)^\beta}{\beta^\beta} \right]^{\frac{1}{1+\beta}}.$$

Define

$$\mathcal{F}(\pi) \equiv \frac{Ry_1 + (1 - \pi)y_2}{1 + \beta(1 - \pi)}.$$

Then

$$\mathcal{F}'(\pi) = \frac{\beta Ry_1 - y_2}{(1 + \beta(1 - \pi))^2} < 0$$

whenever $y_2 > \beta Ry_1$, so $\mathcal{F}$ is strictly decreasing. Moreover,

$$\hat{b}(0) = \frac{y_2 - \beta Ry_1}{1 + \beta} = b^u > \theta_L.$$

Since b^u is the unique maximizer of

$$b \mapsto \log \left(y_1 + \frac{b}{R} \right) + \beta \log(y_2 - b),$$

the high type strictly prefers $\hat{b}(0)$ to safe pooling. Therefore the high-type IC holds strictly at $\pi = 0$. By continuity and monotonicity, there exists a unique cutoff $\hat{\pi}_H \in (0, 1]$ such that the high-type IC holds if and only if

$$\pi \leq \hat{\pi}_H.$$

If

$$\left[\frac{(Ry_1 + \theta_L)(y_2 - \theta_L)^\beta}{\beta^\beta} \right]^{\frac{1}{1+\beta}} > Ry_1,$$

then π_H is characterized by

$$\frac{Ry_1 + (1 - \pi_H)y_2}{1 + \beta(1 - \pi_H)} = \left[\frac{(Ry_1 + \theta_L)(y_2 - \theta_L)^\beta}{\beta^\beta} \right]^{\frac{1}{1+\beta}},$$

that is,

$$\pi_H = 1 - \frac{\left[\frac{(Ry_1 + \theta_L)(y_2 - \theta_L)^\beta}{\beta^\beta} \right]^{\frac{1}{1+\beta}} - Ry_1}{y_2 - \beta \left[\frac{(Ry_1 + \theta_L)(y_2 - \theta_L)^\beta}{\beta^\beta} \right]^{\frac{1}{1+\beta}}}.$$

If instead

$$\left[\frac{(Ry_1 + \theta_L)(y_2 - \theta_L)^\beta}{\beta^\beta} \right]^{\frac{1}{1+\beta}} \leq Ry_1,$$

then $\pi_H = 1$.

Putting the three conditions together, $\hat{b}(\pi)$ can be sustained as a risky-pooling equilibrium whenever

$$\pi \leq \min\{\pi_C, \pi_L, \pi_H\}.$$

Hence, defining

$$\hat{\pi} \equiv \min\{\pi_C, \pi_L, \pi_H\},$$

we have $\hat{\pi} > 0$ under the interesting-region assumption, and for every $\pi \in [0, \hat{\pi}]$,

$$\hat{b}(\pi) \in \mathcal{B}^R(\pi), \quad \bar{V}_{ND}(\pi) = \hat{V}(\hat{b}(\pi); \pi).$$

That is, for sufficiently small π , the ex-ante maximizing risky-pooling debt is itself sustainable as a risky-pooling equilibrium.

FD vs. Upper bound of the ND set

Under full disclosure, the low type is constrained and chooses $b = \theta_L$, while the high type is unconstrained and chooses

$$b = \frac{y_2 - \beta R y_1}{1 + \beta}.$$

Hence the ex-ante full-disclosure payoff is

$$\begin{aligned} V_{FD}(\pi) &= \pi \left[\log \left(y_1 + \frac{\theta_L}{R} \right) + \beta \log(y_2 - \theta_L) \right] \\ &\quad + (1 - \pi) \left[\log \left(y_1 + \frac{1}{R} \frac{y_2 - \beta R y_1}{1 + \beta} \right) + \beta \log \left(y_2 - \frac{y_2 - \beta R y_1}{1 + \beta} \right) \right] \\ &= \pi \left[\log \left(y_1 + \frac{\theta_L}{R} \right) + \beta \log(y_2 - \theta_L) \right] \\ &\quad + (1 - \pi) \left[\log \left(\frac{R y_1 + y_2}{R(1 + \beta)} \right) + \beta \log \left(\frac{\beta(R y_1 + y_2)}{1 + \beta} \right) \right]. \end{aligned}$$

Under non-disclosure, the ex-ante best risky-pooling debt is

$$\hat{b}(\pi) = \frac{y_2 - \beta R y_1}{1 + \beta(1 - \pi)}.$$

Substituting $\hat{b}(\pi)$ into the risky-pooling payoff gives

$$\begin{aligned}\bar{V}_{ND}(\pi) &= \log \left(y_1 + \frac{1 - \pi}{R} \hat{b}(\pi) \right) + \beta \pi \log(y_2 - \theta_L) + \beta(1 - \pi) \log(y_2 - \hat{b}(\pi)) \\ &= \log \left(\frac{R y_1 + (1 - \pi) y_2}{R(1 + \beta(1 - \pi))} \right) + \beta \pi \log(y_2 - \theta_L) \\ &\quad + \beta(1 - \pi) \log \left(\frac{\beta(R y_1 + (1 - \pi) y_2)}{1 + \beta(1 - \pi)} \right).\end{aligned}$$

At $\pi = 0$, we have

$$\hat{b}(0) = \frac{y_2 - \beta R y_1}{1 + \beta},$$

which is exactly the high type's full-disclosure debt choice. Therefore,

$$\begin{aligned}\bar{V}_{\text{ND}}(0) &= \log\left(\frac{Ry_1 + y_2}{R(1 + \beta)}\right) + \beta \log\left(\frac{\beta(Ry_1 + y_2)}{1 + \beta}\right) \\ &= V_{\text{FD}}(0).\end{aligned}$$

Now define

$$\Delta(\pi) \equiv \bar{V}_{\text{ND}}(\pi) - V_{\text{FD}}(\pi).$$

Since $\Delta(0) = 0$, the sign of $\Delta(\pi)$ for small π is governed by $\Delta'(0)$.

Using the envelope theorem,

$$\bar{V}'_{\text{ND}}(\pi) = -\frac{\hat{b}(\pi)/R}{y_1 + \frac{1-\pi}{R}\hat{b}(\pi)} + \beta \log(y_2 - \theta_L) - \beta \log(y_2 - \hat{b}(\pi)).$$

Evaluating at $\pi = 0$,

$$\begin{aligned}\bar{V}'_{\text{ND}}(0) &= -\frac{\frac{1}{R} \frac{y_2 - \beta Ry_1}{1 + \beta}}{y_1 + \frac{1}{R} \frac{y_2 - \beta Ry_1}{1 + \beta}} + \beta \log\left(\frac{y_2 - \theta_L}{y_2 - \frac{y_2 - \beta Ry_1}{1 + \beta}}\right) \\ &= -\frac{y_2 - \beta Ry_1}{Ry_1 + y_2} + \beta \log\left(\frac{(1 + \beta)(y_2 - \theta_L)}{\beta(Ry_1 + y_2)}\right).\end{aligned}$$

Since $V_{\text{FD}}(\pi)$ is linear in π ,

$$\begin{aligned} V'_{\text{FD}}(0) &= \log\left(y_1 + \frac{\theta_L}{R}\right) + \beta \log(y_2 - \theta_L) \\ &\quad - \log\left(\frac{Ry_1 + y_2}{R(1 + \beta)}\right) - \beta \log\left(\frac{\beta(Ry_1 + y_2)}{1 + \beta}\right) \\ &= \log\left(\frac{(1 + \beta)(Ry_1 + \theta_L)}{Ry_1 + y_2}\right) + \beta \log\left(\frac{(1 + \beta)(y_2 - \theta_L)}{\beta(Ry_1 + y_2)}\right). \end{aligned}$$

Subtracting,

$$\begin{aligned} \Delta'(0) &= \bar{V}'_{\text{ND}}(0) - V'_{\text{FD}}(0) \\ &= \log\left(\frac{Ry_1 + y_2}{(1 + \beta)(Ry_1 + \theta_L)}\right) - \frac{y_2 - \beta Ry_1}{Ry_1 + y_2}. \end{aligned}$$

Therefore, $\Delta'(0) > 0$ whenever

$$\log \left(\frac{Ry_1 + y_2}{(1 + \beta)(Ry_1 + \theta_L)} \right) > \frac{y_2 - \beta Ry_1}{Ry_1 + y_2}. \quad (**)$$

Under (**), we have $\Delta(0) = 0$ and $\Delta'(0) > 0$, so by continuity there exists $\varepsilon > 0$ such that

$$\bar{V}_{ND}(\pi) > V_{FD}(\pi) \quad \text{for all } \pi \in (0, \varepsilon).$$

Condition (**) states that, locally around $\pi = 0$, the first-order gain from relaxing the low type's borrowing constraint under pooling dominates the first-order adverse-selection distortion created by non-disclosure.

Discussion of Assumptions

Table: Average Duration in Regime (Years Since Entry)

Regime	n_{spells}	Mean	Median	Std. Dev.	n_{censored}	n_{upgraded}
e-GDDS	109	15.19	15.85	6.21	34	75
e-GDDS+NSDP	75	6.95	7.28	2.23	75	0
SDDS	80	13.25	10.68	12.84	48	32
SDDS Plus	32	7.56	7.66	3.05	32	0

Note: “Censored” spells are ongoing (the country has not yet upgraded). Mean and median include censored spells, so they understate the true expected duration if many spells are still active.

Estimate the prob. of adoption conditional on not having adopted a higher tier.

$$\mathbb{P}(\text{Adopt}_i^r = 1 \mid \text{CouldAdopt}_i^r = 1) = \Lambda(\alpha_i + \gamma Z_{i,t+1} + \Gamma' X_{i,t} + \varepsilon_{it})$$

	SDDS	GDDS	NSDP	Any
Log GDP	0.005 (0.031)	0.451** (0.217)	3.254*** (0.676)	0.019 (0.030)
Reserves / external debt	-0.001 (0.004)	0.011 (0.041)	-0.350 (0.313)	0.000 (0.007)
Short-term debt share	0.003*** (0.001)	-0.011*** (0.004)	0.026 (0.023)	0.001 (0.001)
External debt / GDP	0.036 (0.023)	-0.160 (0.108)	1.502** (0.573)	0.038 (0.039)
Tax revenue / GDP	-0.001 (0.003)	0.005 (0.012)	-0.009 (0.029)	0.006 (0.004)
GDP growth, lead	-0.000 (0.001)	0.003 (0.008)	-0.000 (0.012)	0.001 (0.002)
Reserves, lead	0.019 (0.060)	-0.217 (0.523)	-1.563 (1.252)	-0.104 (0.099)
Tax revenue / GDP, lead	0.001 (0.002)	-0.017 (0.012)	0.039 (0.028)	-0.006 (0.004)
Observations	777	206	129	777
Adopters	13	21	23	35

Note: Linear probability model. Standard errors are in parentheses. The default-cost proxy is the average IMF conditionalities. Lead variables are one-period-ahead fundamentals. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Testable Implications

- ▶ In our model transparency is associated with dispersion of issuances
 - $b_{FD}^*(\theta_L) < b_{FD}^*(\theta_H)$
 - $b_{ND}^*(\theta_L) = b_{FD}^*(\theta_H) = \tilde{b}$
- ▶ Residualize issuances (only countries with no arrears):

$$\log \left(1 + \frac{\text{Issuances}_{it}}{\text{GDP}_{it}} \right) = \alpha_i + \delta_t + \Gamma' X_{it} + u_{it}$$

- ▶ Compare the variance of issuances 5Y after a new tier is launched among groups

$$\mathbb{V}(u_{it} | \text{Transparent}^r = 1) > \mathbb{V}(u_{it} | \text{Transparent}^r = 0)$$

